Module 7: Working with Languages

AWS Academy Natural Language Processing

Module overview

Sections

1. Working with language issues
2. Detecting and translating languages
3. Transcribing and vocalizing text with AWS services

Labs

- Challenge Lab: Implementing a Multilingual Solution

Module objectives

- This module prepares you to do the following:
- Describe the challenges of working with languages
- Identify the predominant language of a text with Amazon Comprehend
- Describe common use cases for Amazon Translate, Amazon Polly, and Amazon Transcribe
- Implement a solution for transcribing and translating text

Section 1: Working with language issues

Module 7: Working with Languages

Challenges: Understanding context

- **Context objectives**
  - Who is speaking?
  - What are they speaking about?
  - How do they feel about the subject?
  - Why do they feel that way?

- **Context analysis**
  - N-grams
  - Noun phrase extraction
  - Theme extraction

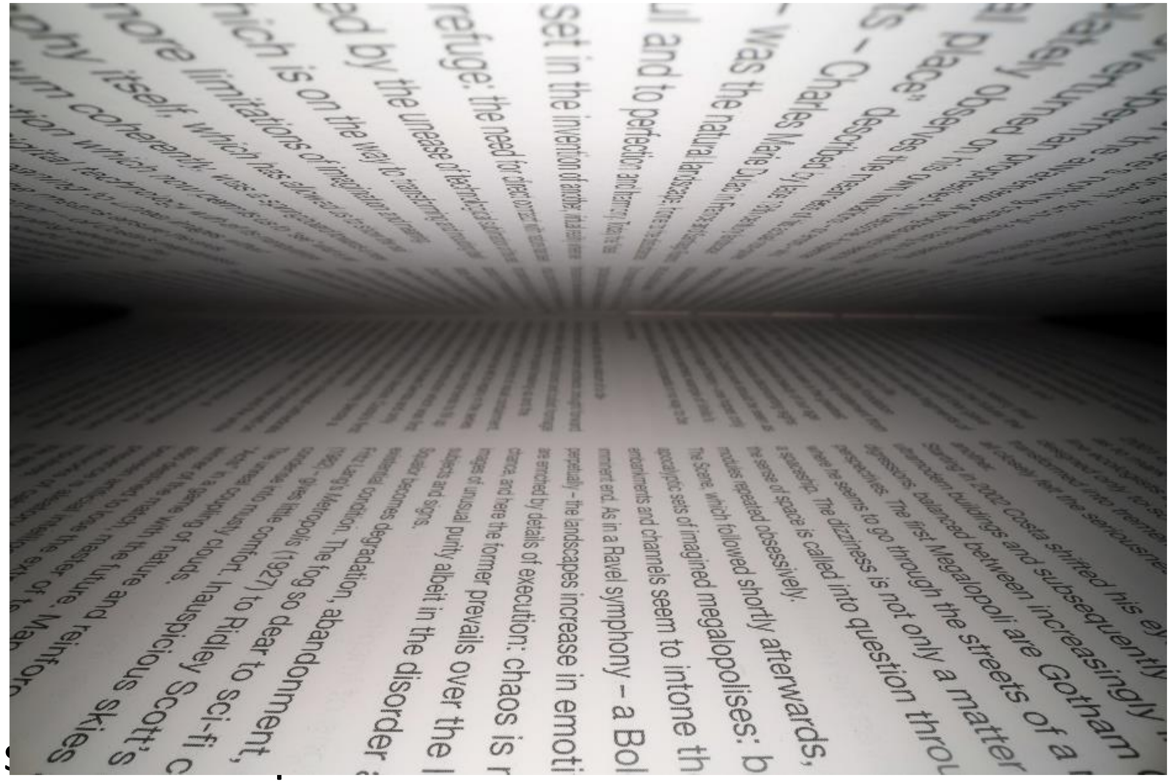
Challenges: Language structure

- Syntax: The way that words are structured in a sentence
  - Word order and directionality
  - Subject, object, verb relationship
  - Possession and plurals
  - Gender agreement for some languages
- Semantic and syntax analysis
  - Semantic parsing
  - Statistical models
- Low-resource languages

English	The dog has black fur.
Spanish	El perro tiene pelaje negro.
English	Look at the dog’s long tail.
German	Schau dir den langen Schwanz des Hundes an.
English	Look at the dog.
English	Look at the dogs.
Albanian	Shiko të gjithë qenin.
Albanian	Shiko të gjithë qentë.

Challenges: Vocabulary and word issues

- Specialized vocabulary
  - Out of vocabulary (OOV) problem: Dealing with words that are not in your working vocabulary
- Homonyms
  - Homograph: *wind* and *wind*
  - Homophone: *hear* vs. *here*
- False cognates
  - English: *sensible -> reasonable*
  - Spanish: *sensible -> sensitive*
- Complex morphologies
  - Turkish: Base words change meaning by adding s



Challenges: Characters and spelling

- Various character sets
  - Chinese-Japanese-Korean (CJK) characters
- Nonstandard spelling
  - Arabic, Hindi
  - Slang, shorthand in texts
- Character embedding
  - Similar to word embedding
  - Groups of letters that are assigned to a vector
    - Similar character vectors are grouped together

Language use cases

Machine translation

Subtitle generation

Automatic transcription

Text analysis

Section 1: Summary

- Language challenges
  - Understanding context
  - Language structure
  - Vocabularies
  - Character sets
  - Machine translations
- Language use cases
  - Transcription
  - Translation
  - Text and voice

Section 2: Detecting and translating languages

Module 7: Working with Languages

Language detection tools

- Amazon Comprehend
  - Uses a pretrained model
  - Can detect more than 100 different languages
- Python packages
  - langdetect
  - spaCy
  - fastText

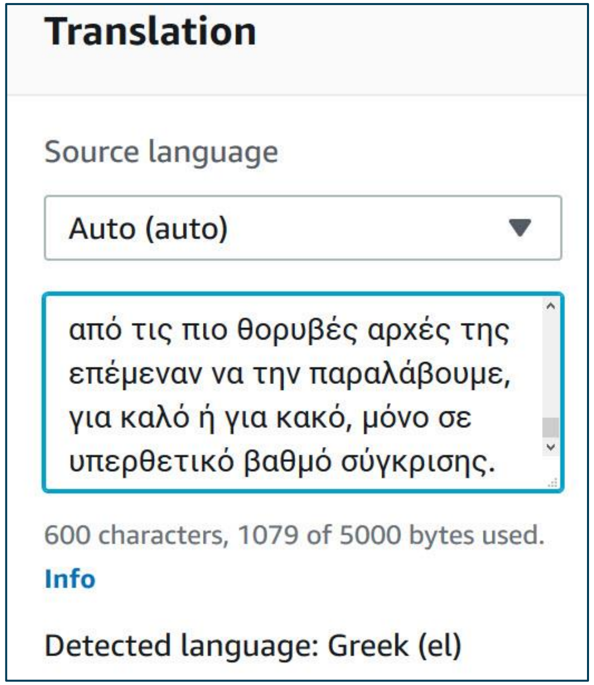
langdetect example

```
pip install langdetect
from langdetect import detect
detect ("this is some text")

Result 'en'
```

Translation tools

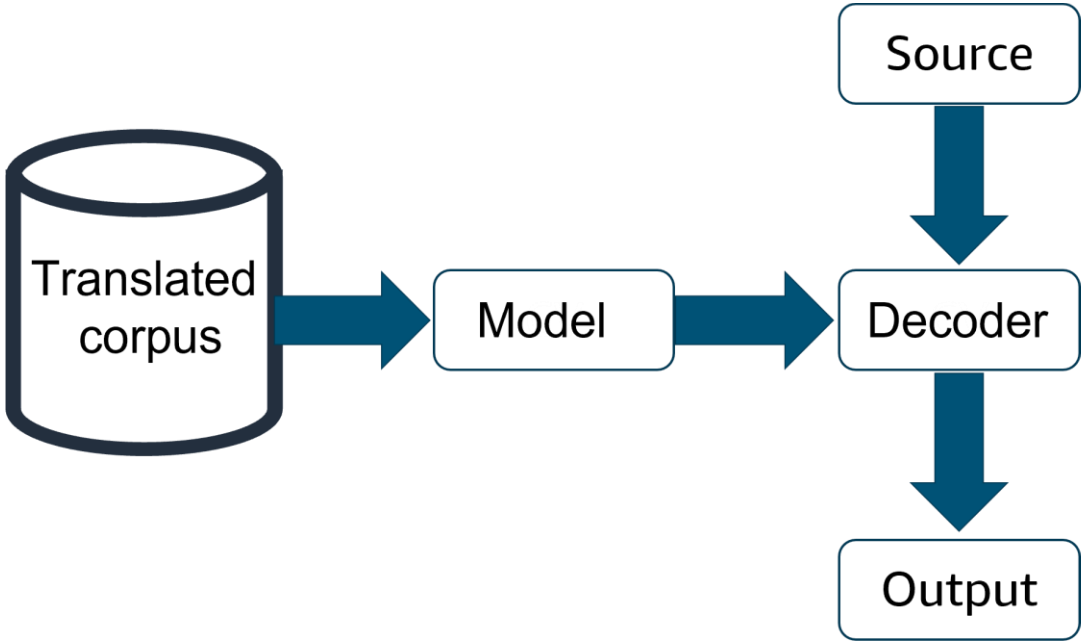
- Amazon Translate
  - Uses a pretrained model
  - Can detect more than 100 different languages
- Python packages
  - langdetect
  - translate
  - fastText




```
import boto3
translate = boto3.client(service_name='translate',
                         region_name='us-east-1', use_ssl=True)
result = translate.translate_text(Text="Hello, World",
              SourceLanguageCode="en", TargetLanguageCode="es")
              print('TranslatedText: ' + result.get('TranslatedText'))
```

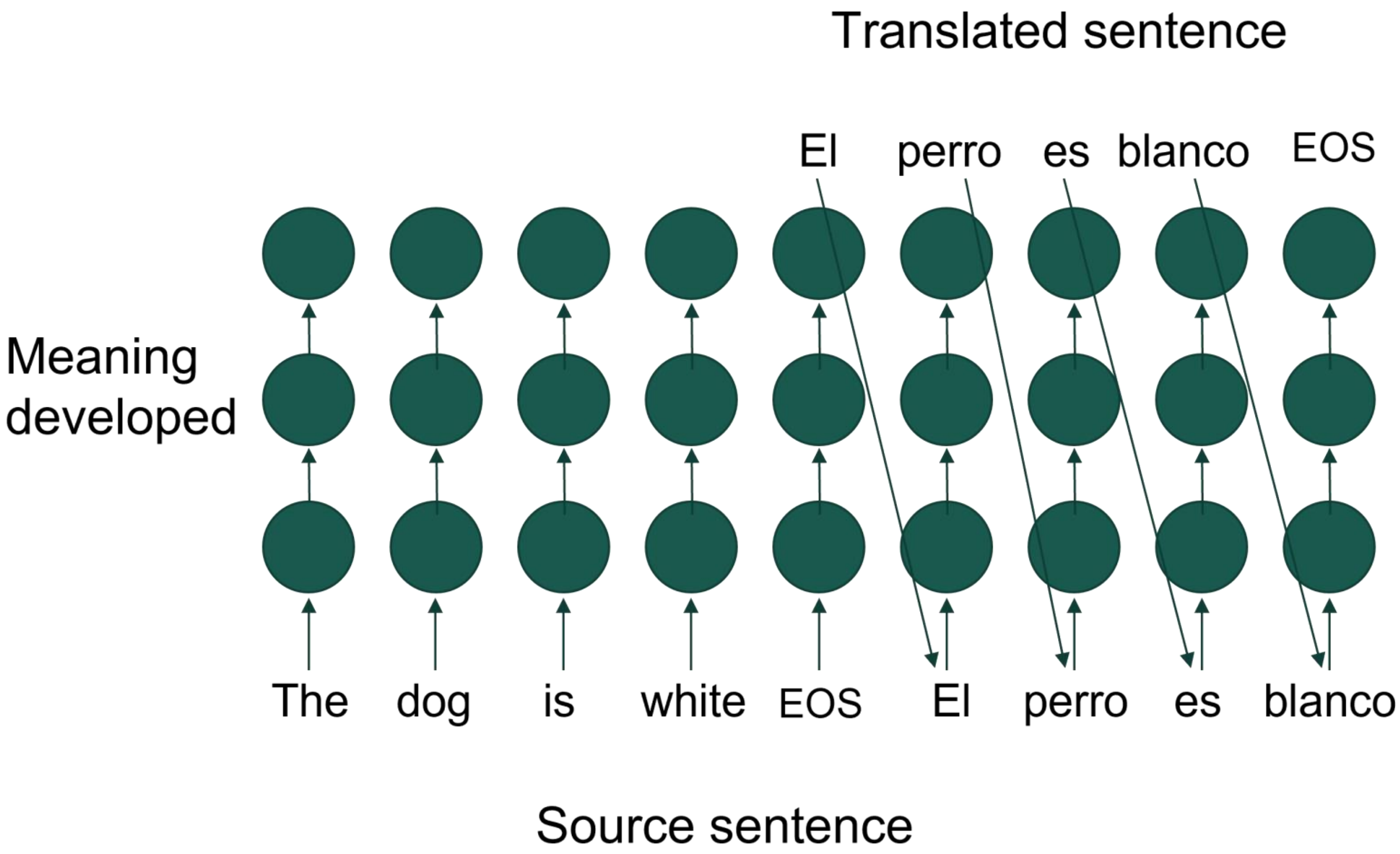
Statistical machine translation (SMT)

- SMT systems are based on applying statistical models to predict translations
- Words and phrases are the units of translation
- Challenges
  - Understanding context
  - Works better for some language pairs
  - Languages with dramatic differences in word order



Neural machine translation (NMT)

- NMT systems use neural networks to learn a model for translation
  - Built with an encoder-decoder model
  - Most use recurrent neural networks
- Challenges
  - Long sequences of text
  - Large vocabularies
  - Long training times
  - Low-resource languages



Section 2: Summary

- Language detection tools
  - Amazon Comprehend
  - Python packages
- Translation tools
  - Amazon Translate
  - Python packages
- Translation methods
  - Statistical machine translation
  - Neural machine translation

Section 3: Transcribing and vocalizing text with AWS services

Module 7: Working with Languages

Transcribing text with Amazon Transcribe

- Convert audio files to text
  - Store files in Amazon Simple Storage Service (Amazon S3)
  - Invoke a job from the console, AWS CLI, or API
- Convert streaming data
  - Real-time translation
  - Invoke text processing from the transcription
- Identify speakers
- Transcribe from multiple channels

{"jobName":"Testtranscribe","accountId":"[redacted]","results":{"transcripts":[{"transcript":"test. Hello? Hello. This is a test test test test."}],"items":[{"start_time":"0.0","end_time":"0.47","alternatives":
[{"confidence":"0.9075","content":"test"}],"type":"pronunciation"},{"alternatives":
[{"confidence":"0.0","content":"."}],"type":"punctuation"},{"start_time":"0.48","end_time":"1.29","alternatives":
[{"confidence":"1.0","content":"Hello"}],"type":"pronunciation"},{"alternatives":
[{"confidence":"0.0","content":"?"}],"type":"punctuation"},{"start_time":"1.3","end_time":"1.79","alternatives":
[{"confidence":"1.0","content":"Hello"}],"type":"pronunciation"},{"alternatives":
[{"confidence":"0.0","content":"."}],"type":"punctuation"},{"start_time":"1.8","end_time":"2.46","alternatives":
[{"confidence":"0.9154","content":"Hello"}],"type":"pronunciation"},{"alternatives":

Sample output

Amazon Transcribe use cases

Speech-to-text notifications

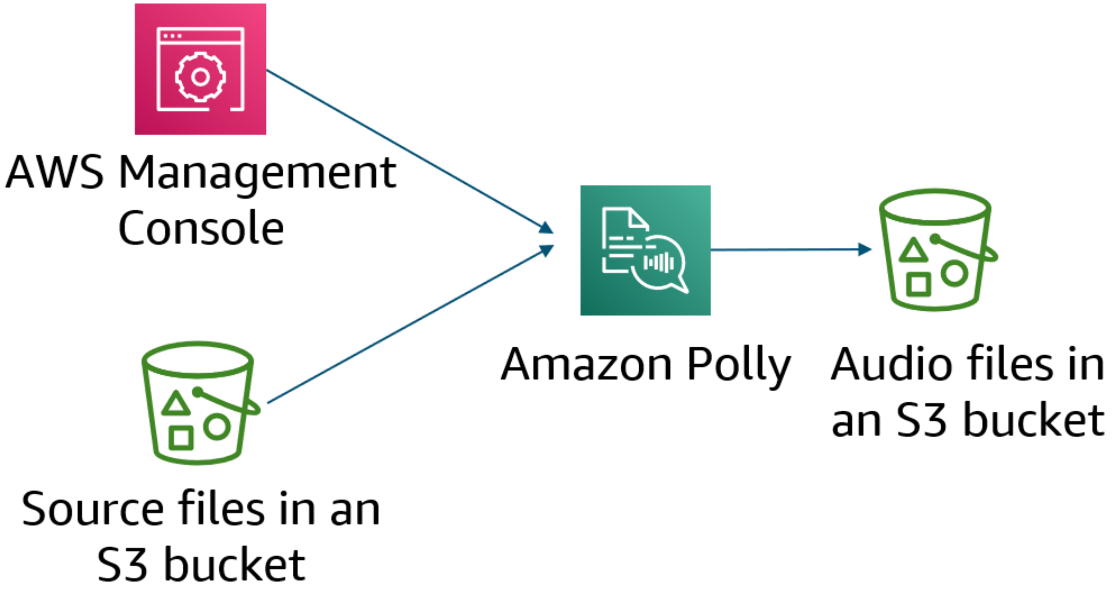
Medical and legal transcriptions

Call center analysis

Real-time translations

Converting text to voice with Amazon Polly

- Input from text files
- Real-time processing or batch processing
- Custom voices
- Neural text-to-speech (TTS) for higher quality output
- Custom lexicons for specialized terminology
- Speech Synthesis Markup Language (SSML) support



Modifying Amazon Polly output

- Common SSML tags

Action	Tag
Add a pause	<break>
Emphasize words	<emphasis>
Control volume, rate, and pitch	<prosody>

<speak>Hi, I'm going to demonstrate some SSML tags.<break> That was a short pause. You can also create longer pauses, like this <break time="3s"/.> With the prosody tag, you can set the volume <prosody value="loud">like this.</prosody></speak>

- Speech marks
  - Synchronize speech with video
- Custom lexicons

Amazon Polly use cases

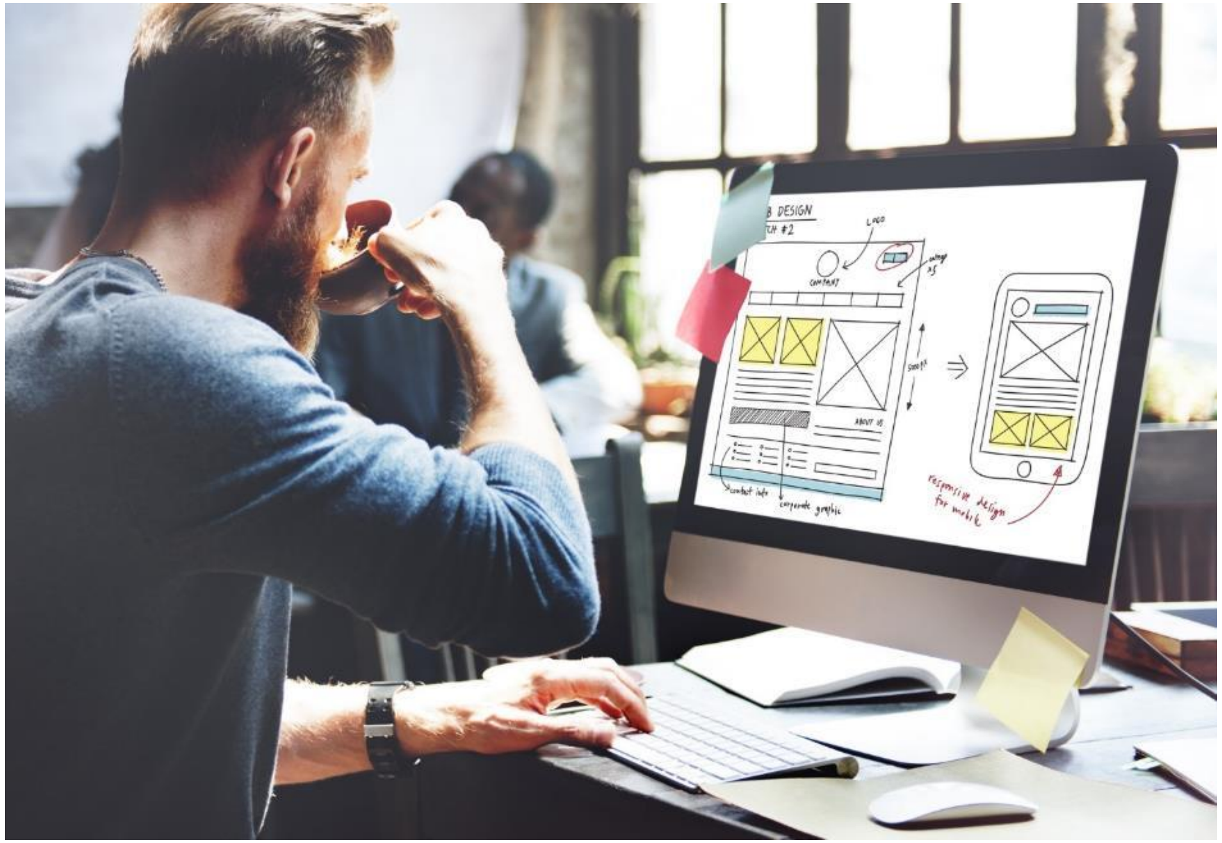
Audio on websites

Voice response
for contact centers

Animations with speech

Vocalized alerting

Module 7 – Challenge Lab 1: Implementing a Multilingual Solution



Section 3: Summary

- Amazon Transcribe
  - Overview
  - Use cases
- Amazon Polly
  - Overview
  - Use cases

Thank you

Corrections, feedback, or other questions?
Contact us at https://support.aws.amazon.com/#/contacts/aws-academy.